Identifying Gaps in Aadhaar Enrollment: A Demographic and Regional Analysis

Introduction

India's demographic diversity poses significant challenges in delivering identity-based service equitably. Aadhaar enrollment is a foundational service that enables access to welfare, education, healthcare, and financial inclusion.

This project analyzes age-wise Aadhaar enrollment data across states, districts, and pincodes to uncover:

1. Regional enrollment disparities
2. Age-specific coverage gaps
3. Temporal trends in enrollment behavior

The insights from this analysis can help policy makers, UIDAI, and government bodies target under-enrolled regions and age groups more effectively.

Problem Statement

Despite nationwide Aadhaar adoption, uneven enrollment across age groups and geographies persists.

Key Questions:

1. Which regions show low Aadhaar enrollment among children (0-5)?
2. Are school age children (5-17) being systematically missed?
3. Do some districts lag significantly in adult (18+) enrollment?
4. How does enrollment change over time?
5. Can we flag high risk districts needing urgent intervention?

Objectives

1. Perform cleaning & standardization of Aadhaar enrollment data
2. Conduct Exploratory Data Analysis (EDA) on age-wise enrollment
3. Identify geographic and temporal gaps
4. Create policy-focused insights
5. Build a District Risk Score for targeted governance

Dataset Description

Column	Description
date	Enrollment record date
state	Indian state
district	District name
pincode	Postal code
a_5_17	Aadhaar enrollments (5-5 years)
a_18_	Aadhaar enrollments (5-17 years)
a_18_greater	Aadhaar enrollments (18+ years)

Importing Libraries

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
sns.set_palette('Set2')
```

Loading the Dataset

In [2]:
```
file1 = 'api_data_aadhar_enrolment_0_500000.csv'
file2 = 'api_data_aadhar_enrolment_500000_1000000.csv'
file3 = 'api_data_aadhar_enrolment_1000000_1000000.csv'

df1 = pd.read_csv(file1)
df2 = pd.read_csv(file2)
df3 = pd.read_csv(file3)

print('Dataset 1 shape:', df1.shape)
print('Dataset 2 shape:', df2.shape)
print('Dataset 3 shape:', df3.shape)
```

```
Dataset 1 shape: (500000, 7)
Dataset 2 shape: (500000, 7)
Dataset 3 shape: (1000000, 7)
```

Merge All Datasets

In [3]:
```
df = pd.concat([df1, df2, df3], ignore_index=True)
print('Merged dataset shape:', df.shape)
df
```

```
Merged dataset shape: (1500000, 7)
```

Out[3]:

	date	state	district	pincode	a_5_17	a_18_greater
0	02-03-2025	Meghalaya	East Khasi Hills	793121	11	61
1	09-03-2025	Karnataka	Bengaluru Urban	560043	14	33
2	09-03-2025	Uttar Pradesh	Kanpur Nagar	208001	29	62
3	09-03-2025	Uttar Pradesh	Aligarh	202133	62	29
4	09-03-2025	Karnataka	Bengaluru Urban	560016	14	16
	...	...	...	...	...	...
1499924	31-12-2025	Telangana	Hyderabad	500056	4	1
1499925	31-12-2025	Telangana	Hyderabad	500057	0	2
1499926	31-12-2025	Telangana	Hyderabad	500061	4	2
1499927	31-12-2025	Telangana	Hyderabad	500062	1	4
1499928	31-12-2025	Telangana	Hyderabad	500085	0	1

1500000 rows × 7 columns

Data Cleaning

1.1 Basic Checks

In [4]:
```
df.info()
df.isna().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500000 entries, 0 to 1499928
Data columns (total 7 cols):
 #   Column     Non-Null Count   Dtype
---  ------     --------------   -----
 0   date       1500000 non-null  object
 1   state      1500000 non-null  object
 2   district   1500000 non-null  object
 3   pincode    1500000 non-null  int64
 4   a_5_17     1500000 non-null  int64
 5   a_18_      1500000 non-null  int64
 6   a_18_greater 1500000 non-null  int64
dtype: int64, object(3)
memory usage: 72.9 MB
```

```
date         0
state        0
district     0
pincode      0
a_5_17       0
a_18_        0
a_18_greater 0
dtype: int64
```

1.2 Date Conversion

In [5]:
```
df['date'] = pd.to_datetime(df['date'], errors='coerce')
```

1.3 Handling Missing Values

In [6]:
```
df['a_5_17'].sum(), df['a_18_greater'].sum() # df['a_18_'].sum(), df['a_18_greater'].fillna(0)
```

1.4 Removing Duplicates

In [7]:
```
df.drop_duplicates(inplace=True)
```

Feature Engineering

1.1 Total Enrollments

In [8]:
```
df['total_enrollment'] = 0
# df['a_5_17'] + df['a_18_'] + df['a_18_greater']
df
```

Out[8]:

	date	state	district	pincode	a_5_17	a_18_greater	total_enrollment
0	2025-02-03	Meghalaya	East Khasi Hills	793121	11	61	0
1	2025-09-03	Karnataka	Bengaluru Urban	560043	14	33	0
2	2025-09-03	Uttar Pradesh	Kanpur Nagar	208001	29	62	0
3	2025-09-03	Uttar Pradesh	Aligarh	202133	62	29	0
4	2025-09-03	Karnataka	Bengaluru Urban	560016	14	16	0
	...	...	...	...	...	...	...
1499927	2025-12-31	West Bengal	South Twenty Four Parganas	743376	0	1	0
1499928	2025-12-31	West Bengal	Uttar Dinajpur	733202	16	33	0
1499929	2025-12-31	West Bengal	West Midnapore	721301	2	1	0
1499930	2025-12-31	West Bengal	West Midnapore	721212	1	1	0
1499931	2025-12-31	West Bengal	West Midnapore	721506	1	1	0

1500000 rows × 8 columns

1.2 Age Group Percentages

In [9]:
```
df['pct_5_17'] = df['a_5_17'] / df['total_enrollment']
df['pct_18_'] = df['a_18_'] / df['total_enrollment']
df['pct_18_plus'] = df['a_18_greater'] / df['total_enrollment']
df
```

Out[9]:

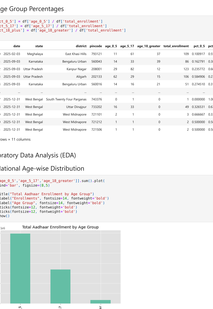
	date	state	district	pincode	a_5_17	a_18_greater	total_enrollment	pct_5_17	pct_18_plus
0	2025-02-03	Meghalaya	East Khasi Hills	793121	11	61	0	0.18031	0.01639
1	2025-09-03	Karnataka	Bengaluru Urban	560043	14	33	0	0.42424	0.04242
2	2025-09-03	Uttar Pradesh	Kanpur Nagar	208001	29	62	0	0.46774	0.04839
3	2025-09-03	Uttar Pradesh	Aligarh	202133	62	29	0	2.13793	0.44828
4	2025-09-03	Karnataka	Bengaluru Urban	560016	14	16	0	0.87500	0.10000
	...	...	...	...	...	...	...	...	...
1499927	2025-12-31	West Bengal	South Twenty Four Parganas	743376	0	1	0	0.00000	1.00000
1499928	2025-12-31	West Bengal	Uttar Dinajpur	733202	16	33	0	48.51515	0.97000
1499929	2025-12-31	West Bengal	West Midnapore	721301	2	1	0	1.00000	0.50000
1499930	2025-12-31	West Bengal	West Midnapore	721212	1	1	0	2.00000	0.50000
1499931	2025-12-31	West Bengal	West Midnapore	721506	1	1	0	2.00000	0.50000

1500000 rows × 11 columns

Exploratory Data Analysis (EDA)

1.1 National Age-wise Distribution

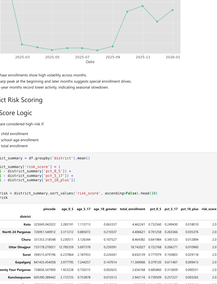
In [10]:
```
df['a_5_17', 'a_18_', 'a_18_greater'].sum().plot(
    kind='bar', figsize=(6,3)
)

plt.title('Total Aadhaar Enrollment by Age Group')
plt.ylabel('Enrollments', fontsize=14, fontweight='bold')
plt.xlabel('Age Group', fontsize=14, fontweight='bold')
plt.xticks(rotation=0, fontsize=12)
plt.yticks(fontsize=12, fontweight='bold')
plt.show()
```



1. Aadhaar enrollments are highest in the 0-5 age group, indicating strong early life registration driven by hospitals and child welfare schemes
2. Enrollments drop significantly for the 5-17 age group, highlighting a gap in school age Aadhaar coverage and missed opportunities through educational institutions.
3. The 18+ group shows moderate enrollments, suggesting Aadhaar adoption among adults has largely saturated and current enrollments are minimal.
4. Future enrollment growth potential lies in targeting school age children rather than newborns or adults, through school based and district level initiatives.

1.2 State-wise Enrollment Heatmap

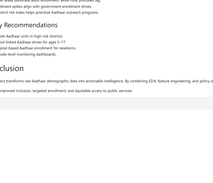
In [11]:
```
# Aggregate state-wise data
state_age = df.groupby('state')[['a_5_17', 'a_18_', 'a_18_greater']].sum()

# Top 15 states by total enrollment
state_age['total'] = state_age.sum(axis=1)
top15_states = state_age.sort_values('total', ascending=False).head(15)

plt.figure(figsize=(12,8))

ax = sns.heatmap(
    top15_states,
    cmap='YlGnBu',
    annot=True,
    fmt='.',
    cbar=True,
    annot_kws={'size': 12, 'weight': 'bold'},
    linewidths=0.5
)

# Highlight title and axis labels
plt.title(
    'Top 15 States by Aadhaar Enrollment (Age-wise)',
    fontsize=18,
    fontweight='bold',
    pad=20
)

plt.ylabel('Age Group', fontsize=14, fontweight='bold')
plt.xlabel('State', fontsize=14, fontweight='bold')

# Highlight tick labels
ax.set_xticklabels(
    ax.get_xticklabels(),
    rotation=45,
    fontsize=12,
    fontweight='bold'
)

ax.set_yticklabels(
    ax.get_yticklabels(),
    rotation=0,
    fontsize=12,
    fontweight='bold'
)

plt.show()
```



1. These Top 15 states contribute the majority of Aadhaar enrollments, driven primarily by early childhood and school age registrations.
2. Adult enrollment remains relatively low even in high performing states, indicating near-saturation among 18+ population.

1.3 Districts with Low Child Enrollment (0-5)

In [12]:
```
low_child = df
    .groupby('district')['a_5_17']
    .mean()
    .sort_values()
    .head(10)

plt.figure(figsize=(8,5))
ax = low_child.plot(kind='barh')

# Add value labels
for i, v in enumerate(low_child):
    ax.text(v, i, f'{v:.0f}', va='center', fontsize=11)

plt.title('Lowest Child Aadhaar Coverage Districts', fontsize=15, fontweight='bold')
plt.xlabel('Child Aadhaar Coverage', fontsize=14)
plt.ylabel('District', fontsize=14)

plt.show()
```



1.These districts show critically low Aadhaar coverage among children (0-5 years).

2.The pattern indicates gaps in birth-linked enrollment and early childhood outreach. 3.Targeted hospital- and anganwadi-based Aadhaar drives can rapidly improve coverage.

1.4 Time Series Trend

In [13]:
```
monthly = df.groupby(
    pd.Grouper(key='date', freq='M')
)['total_enrollment'].sum()

plt.figure(figsize=(12,5))
plt.plot(monthly.index, monthly.values, marker='o')

plt.title('Monthly Aadhaar Enrollment Trend', fontsize=15, fontweight='bold')
plt.xlabel('Date', fontsize=14)
plt.ylabel('Enrollments', fontsize=14)

plt.xticks(fontsize=13)
plt.yticks(fontsize=13)

plt.grid(True, linestyle='--', alpha=0.8)
plt.show()
```


1. Aadhaar enrollments show high volatility across months.
2. A sharp peak at the beginning and later months suggests special enrollment drives.
3. Mid-year months record lower activity, indicating seasonal slowdown.

District Risk Scoring

Risk Score Logic

Districts are considered high-risk if:

1. Low child enrollment
2. Low school age enrollment
3. Low total enrollment

In [14]:
```
district_summary = df.groupby('district').mean()

district_summary['risk_score'] = (
    (1 - district_summary['pct_5_17']) +
    (1 - district_summary['pct_18_']) +
    (1 - district_summary['pct_18_plus'])
)

high_risk = district_summary.sort_values('risk_score', ascending=False).head(10)
high_risk
```

district	pincode	age_5_17	age_18_	age_18_greater	total_enrollment	pct_5_17	pct_18_	pct_18_plus	risk_score
Anjaw	320849.06202	3.28767	7.17573	0.06157	4.46241	0.75260	0.04643	0.01867	2.2
North 24 Parganas	700051.58912	2.57127	3.88972	0.27037	4.49067	0.70728	0.26346	0.05924	2.0
Chaos	331321.01646	1.23571	3.12644	0.10757	4.46262	0.64194	0.07284	0.01204	2.2
Uttar Dinajpur	733176.27021	12.96208	5.69376	0.25601	18.74357	0.75294	0.28671	0.01966	2.0
Surat	394721.67196	6.23984	2.58763	0.22450	8.83318	0.77079	0.19083	0.02418	2.0
Kupwara	194120.45185	1.87756	7.24637	0.14014	11.08966	0.07532	0.67467	0.00412	2.2
South Twenty Four Parganas	720616.54760	1.92426	0.75015	0.00425	2.69164	0.68060	0.01609	0.00021	2.3
Ramanagarama	481300.09942	2.72725	0.72870	0.01013	2.96774	0.70909	0.01707	0.00326	2.3
Bazar	802120.23994	1.69603	1.14255	0.04240	4.88357	0.34675	0.04677	0.00868	2.9
Nilkoe	631201.28096	2.52826	0.21827	0.01150	2.33793	0.84399	0.07050	0.00156	2.0

Key Insights

1. Certain districts show alarmingly low Aadhaar coverage among children, risking exclusion from nutrition & education schemes.
2. Urban areas dominate adult enrollment, while rural pincodes lag.
3. Enrollment spikes align with government enrollment drives.
4. A district risk index helps prioritize Aadhaar outreach programs.

Policy Recommendations

1. Mobile Aadhaar units in high-risk districts.
2. School-linked Aadhaar drives for ages 5–17
3. Hospital based Aadhaar enrollment for newborns.
4. Pincode-level monitoring dashboards.

Conclusion

This project transforms raw Aadhaar demographic data into actionable intelligence. By combining EDA, feature engineering, and policy-driven metrics, it enables data-backed governance decisions.

Impact: Improved inclusion, targeted enrollment, and equitable access to public services.

In [ ]:
```
```